

SAURABH BHAGCHANDANI

Geospatial Analyst | Remote Sensing Specialist | WebGIS

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

OVERVIEW

Geospatial analyst with a strong foundation in Geography, applying spatial thinking to understand how location, context and relationships shape patterns within data, communicating spatial insights through clear visual outputs, and using them to support planning, evidence-based decisions and real-world problem-solving.

TECHNICAL SKILLS

GIS & Spatial Analysis: ArcGIS Pro, QGIS, spatial analysis, spatial modelling, spatial statistics, cartography & geospatial visualisation, coordinate reference systems & transformations, geospatial data quality & validation

Remote Sensing & Earth Observation: Google Earth Engine, satellite imagery processing & interpretation, image preprocessing, image classification, change detection, accuracy assessment & validation, EO data workflows

WebGIS & Geospatial Services: MapLibre, Mapbox, Leaflet, CesiumJS, Streamlit, FastAPI, OGC APIs, WMS, WFS, GeoJSON, web map tiles, interactive web mapping, geospatial dashboard development

Programming & Data Technologies: Python, SQL, R, JavaScript, HTML, CSS, Jupyter Notebook, SPSS

EXPERIENCE

GIS Intern | Geonovum, Amersfoort, Netherlands | *Mar 2026 - May 2026*

- Designed and developed a WebGIS application with a custom frontend and FastAPI backend, integrating BAG data through OGC APIs with CBS basemaps and administrative boundary datasets.
- Implemented multi-level spatial navigation from province to neighbourhood scale, providing structured access to multiple BAG object types and related geographic information.
- Developed interactive mapping and reporting functionality to support spatial exploration, interpretation and practical use of Dutch base-registry data.

Platform: [Dutch Base Registries Dashboard](#)

Student Intern | Jal Smruti, New Delhi, India | *Oct 2024 - Nov 2024*

- Researched urban water-conservation practices, supported household water-scorecard assessments, and documented case studies used in community awareness activities.
- Supported community engagement and awareness activities by translating research findings into practical water-conservation guidance and workshop content.

Undergraduate Researcher | Department of Geography, Dyal Singh College, University of Delhi, New Delhi, India | *Jan 2023 - May 2023*

- Conducted field research on gender disparities in education, health and household decision-making across urban and rural communities in Jodhpur, using questionnaire-based primary data collection.
- Applied sampling and statistical methods to analyse survey data and produced reports and visualisations to communicate the findings.

EDUCATION

MSc Geo-Information Science and Earth Observation | Faculty ITC, University of Twente, Enschede, Netherlands | *Sep 2024 - Jul 2026*

Specialisation: Urban Planning and Management

BA (Hons) Geography | Dyal Singh College, University of Delhi, New Delhi, India | *Jun 2021 - Jun 2024*

GEOSPATIAL PROJECTS

Surface Urban Heat Island Analysis & Spatial Modelling | *MSc Thesis, Faculty ITC, University of Twente*

- Analysed daytime Surface Urban Heat Island patterns across Essen, Wuppertal and Soest using ECOSTRESS LST, Local Climate Zones and environmental indicators.
- Evaluated urban-morphology and environmental drivers using spatial analysis, hotspot analysis and machine-learning modelling.
- Developed an interactive CesiumJS dashboard to visualise LST, SUHI, hotspots, environmental indices, LCZ and 3D urban information.

Links: [Dashboard](#) | [GitHub](#) | [Thesis](#)

Malaria Risk Mapping | *Faculty ITC, University of Twente | Rwanda | Mar 2025*

- Evaluated spatial patterns of malaria hazard and vulnerability in QGIS by integrating environmental, demographic and location-based risk factors.
- Combined spatial analysis with R and SPSS-based statistical analysis to produce integrated risk and hotspot maps for identifying areas of elevated malaria exposure.

Slum Identification & Spatial Analysis | *Faculty ITC, University of Twente | Rio de Janeiro, Brazil & Nairobi, Kenya | Nov 2024*

- Assessed informal settlement patterns in QGIS using remote sensing-derived indicators including terrain, vegetation, built-up density, proximity and accessibility.
- Compared settlement morphology, road-network structure and environmental conditions across Rio de Janeiro and Nairobi to identify spatial characteristics associated with informal development.

TRAINING & CERTIFICATIONS

NL Space Summer School - Netherlands Space Campus, Jun 2025

GIS and Remote Sensing for Geospatial Solutions - Faculty ITC, University of Twente, Sep-Oct 2024

Nature-based Solutions for Disaster and Climate Resilience - UNEP / Partnership for Environment and Disaster Risk Reduction, Jul-Aug 2023

Workshop on Application of Geomatics in Urban Disasters - Indian Institute of Remote Sensing, May 2023

LEADERSHIP & ENGAGEMENT

Assistant Secretary / Media & Communications Coordinator | ORBIT, ITC Student Association, University of Twente | 2025

Coordinated media and communications activities and supported association administration within an international student community.

Additional volunteering: Rotaract Club, Dyal Singh College | Environmental Club, Dyal Singh College | Leo Club Eternity Services

LANGUAGES

English: Full professional proficiency | Hindi: Native | Sindhi: Native | Dutch: Elementary